

Duhme — Architecture & Showcase

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Duhme is an engine for methodically tracking how we know what we claim to know.

It explicitly maps what is observed, what is assumed, and what is contradicted. Humans change much less quickly than Als do, which allows our reasoning logic to operate as a deliberately closed universe. As a bounded epistemic domain, it is not a generic oracle.

We optimize for structural competence within our defined scope rather than overusing LLMs to generate ungrounded certainty. Duhme's universe is deliberately narrow.

Compared to a foundation model predicting the entire internet, the scope is strictly limited. Compared to enterprise systems that model human reality using a user_id and a lead_status dropdown, the ontology is deep. We restrict inference to 80+ canonical human mechanisms and context primitives.

If a concept is not in this structural ontology, a model is not permitted to invent it. The execution substrate maps a specific lexicon of human reality:

Execution Substrate & Structural Lexicon

- **Social & Factional Topology:** public/private divergence, hidden coalitions, succession blocs, faction alignment, informal gatekeeping, decision bypass, proxy conflicts, narrative propagation, social isolation, upward/downward behavioral shifts.
- **Situational Leverage:** actual vs. perceived capability, deterrence, organizational bargaining power, coalition mobilization, resource control, threat display, constraint exploitation, information asymmetry.
- **Defensive & Competitive Mechanisms:** shame, status threat, embarrassment, humiliation, fear, resentment, jealousy, loyalty/obligation conflict, cognitive dissonance, self-justification, motivated reasoning, face-saving, blame externalization, moral disengagement, retaliation, reactance, self-handicapping.
- **Affiliative & Reparative Mechanisms:** compassion, protective concern, gratitude, reciprocity, trust/delegation, attachment, forgiveness, principled moral courage, intrinsic mastery, restitution.

Execution Substrate (Continued)

- **Affect, Drive & Desire:** romantic attraction, perceived desirability, persistent preoccupation, impulse inhibition, and several other related interpersonal dynamics.
- **Cognitive State & Memory:** bounded theory-of-mind (Level 0 through Level 2), unobserved intervals, momentary attention capture, immediate gist representation, memory accessibility, resurfacing cues, remembered vs. historical representation, asymmetric cognitive persistence.
- **Epistemic Reality:** independent evidence paths, shared-origin gossip, contradicting dependencies, missing counterevidence, unobserved coordination channels, reflexive intervention contamination.

We do not ask LLMs to generate open-ended behavioral analysis.

Instead, we leverage the fluent pattern recognition of powerful frontier models and localized engines as structured semantic compilers to map messy human input onto this explicit matrix.

Individual models operate as measurable, performance-tracked components within the reasoning pipeline—whether deployed across a heterogeneous multi-model stack or anchored entirely to a single customer-managed BYOK key.